Telecommunications Policy 38 (2014) 96–104

Contents lists available at ScienceDirect

Telecommunications Policy

URL: www.elsevier.com/locate/telpol

Estimating broadband diffusion in the EU using NUTS 1 regional data

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ARTICLE INFO

Available online 27 September 2013

Keywords:
Broadband diffusion
Inter-platform competition
Intra-platform competition
LLU prices

ABSTRACT

In this paper we use panel data on NUTS 1 regional data for 27 EU countries in the years 2006–2010 to analyze determinants of broadband diffusion. We estimate both linear demand specification and the logistic diffusion function. We find that, after controlling for regional differences due to socioeconomic factors, inter-platform competition approximated by an inter-platform Herfindahl index has a significant positive impact on broadband diffusion. Broadband deployment is lower in countries in which DSL has a greater share in Internet access and it is higher in countries in which cable modem has a greater share in Internet access. Moreover, we find that competition between DSL providers has a significant and positive impact on broadband penetration. First, higher prices for a fully unbundled local loop connection, which represent the cost of providing copper-based Internet services, have a significant and negative impact on broadband penetration. Second, a greater incumbent share in DSL connections has a significant and negative impact on broadband penetration.

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1. Introduction

In the last decade the European Commission has been pursuing the objective of universal broadband access to the Internet according to which all European households should have access to basic broadband by 2013. Moreover, as set out in the Digital Agenda for Europe, "by 2020, all Europeans should have access to Internet of above 30 Megabits per second (Mb/s) and 50% or more of European households have subscriptions above 100Mb/s."[1] The Commission recognizes that it is necessary to develop a comprehensive policy towards broadband deployment based on a mix of technologies and to monitor the progress. The objective of this paper is to provide empirical evidence on the determinants of broadband penetration which should help to guide policies towards achieving these objectives.

In 2011, about 67% of households in the European Union (EU) had fixed broadband access but there is a large variation in adoption rates across the EU countries and across regions within particular countries. The highest broadband penetration could be observed in regions in Sweden and Netherlands with about 85% of households having access to broadband and the lowest in regions in Romania and Bulgaria with about 20% of households having broadband access. There are also differences in the type of technologies which are adopted across countries. Digital Subscriber Line (DSL) is still the most

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[1] Digital Agenda for Europe is a flagship initiative of the Europe 2020 strategy for a smart, sustainable and inclusive economy. See "A Digital Agenda for Europe – COM(2010) 245".

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http://dx.doi.org/10.1016/j.telpol.2013.05.006

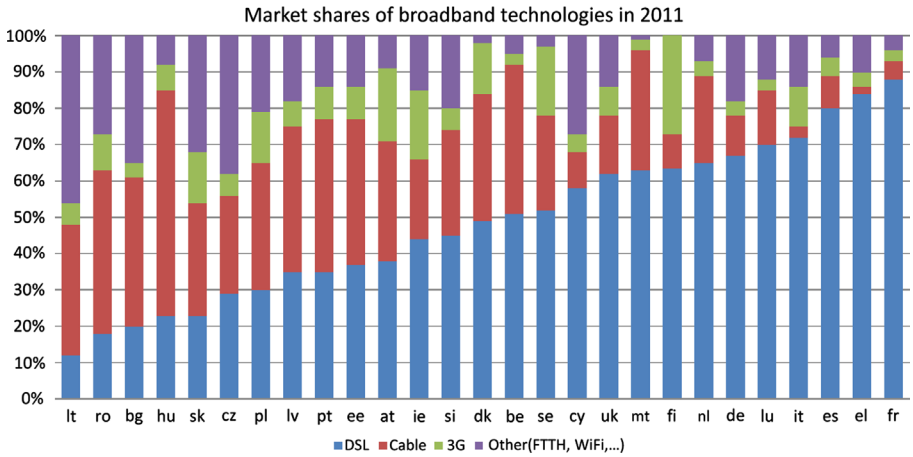

Fig. 1. Market shares of broadband technologies in 2011.
"Eurobarometer: E-Communications Household Surveys", fieldwork in February–March 2011.

commonly used technology in the EU with about 71% of fixed broadband connections as of 2011, but Internet access based on cable modem and fibre to the home (FTTH) has been steadily growing. Also, mobile broadband via modem has been increasingly popular with penetration rate reaching about 8% of the European households as of 2011.[2]

Even though there is a common agenda towards broadband deployment set out by the Commission for all the EU countries, there are significant differences in how broadband adoption evolves due to historical reasons, socioeconomic factors and country-specific approaches towards implementation of policies set out by the Commission. In particular, there are significant differences with respect to the intensity of intra- and inter-platform competition between Western European (WE) and Central and Eastern European countries, where intra-platform competition refers to competition within the DSL platform, and inter-platform competition refers to competition between DSL and other technologies, such as cable modem, FTTH, mobile broadband (see Bouckaert, van Dijk, & Verboven, 2010). In general, there is more inter-platform competition in the new EU countries and more intra-platform competition in the old EU countries. Fig. 1 illustrates shares of broadband technologies in the EU countries as of 2011 and Fig. 2 illustrates incumbent share in DSL connections as of 2010. Similarly, there are differences in the implementation of regulations which may have an impact on broadband diffusion, such as timing and speed of local loop unbundling (LLU) and regulation of prices for fully unbundled local loop connections or shared access.[3]

There is a growing body of studies which analyze determinants of broadband penetration focusing on variables such as intra- and inter-platform competition, LLU price regulation, etc. However, these studies rely mainly on country-level panel data for OECD countries and as such cannot fully capture the degree of heterogeneity which is present on the regional level within the country. We improve upon this by using newly available regional data for the EU countries which is termed the NUTS 1 level data. NUTS is a geocode standard for referencing the administrative divisions of the EU countries for statistical purposes. There are three NUTS levels but due to smaller size not every country has all division levels. For instance, in the case of Luxembourg all three NUTS divisions represent the whole country. We use data at the NUTS 1 level, which refers to basic regions with population ranging between 3 and 7 million inhabitants.[4] There are 97 NUTS 1 regions in 27 EU countries out of which 96 are used in our analysis. Using NUTS 1 level data, after controlling for a number of region-specific socioeconomic variables, we should be able to disentangle the effects of country-level policies on broadband penetration more accurately than in country-level studies.

We estimate both linear demand specification and the logistic diffusion function for NUTS 1 level data in the years 2006–2010. We find that, after controlling for regional differences due to socioeconomic factors, inter-platform competition approximated by inter-platform Herfindahl index has a significant positive impact on broadband diffusion. Broadband deployment is lower in countries in which DSL has a greater share in Internet access and it is higher in countries in which cable modem has a greater share in Internet access. Moreover, we find that competition between DSL providers also has a significant impact on broadband penetration. First, higher prices for a fully unbundled local loop connection, which represent the cost of providing copper-based Internet services, have a significant and negative impact on broadband penetration. Second, a greater incumbent share in DSL connections has a significant and negative impact on broadband penetration.

[2] Source: "Eurobarometer: E-Communications Household Surveys", fieldwork in February–March 2011.

[3] Local loop unbundling is the regulatory process of allowing multiple telecommunications operators to use connections from the telephone exchange to the customer's premises.

[4] Source: http://epp.eurostat.ec.europa.eu/portal/page/portal/nuts_nomenclature/introduction.

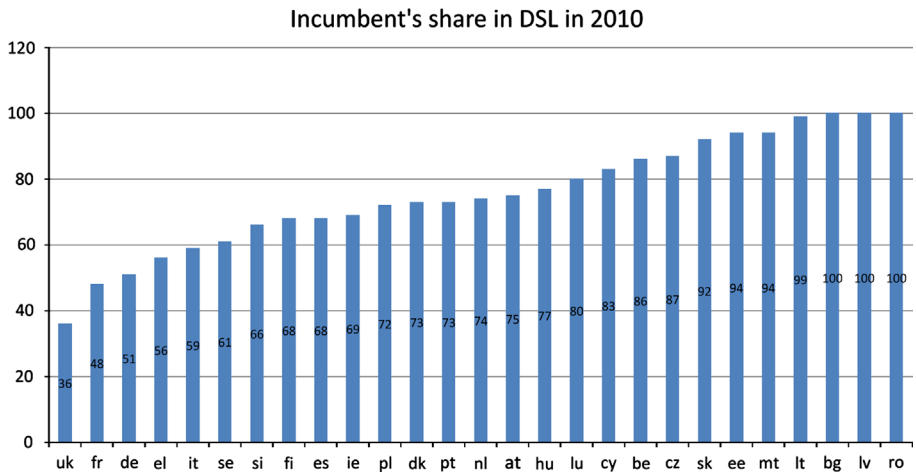

Fig. 2. Incumbent share in DSL connections in 2010.
"15th Progress Reports on the Single European Electronic Communications Market", August 2010.

The remainder of the paper is organized as follows. Section 2 summarizes previous research on broadband diffusion. Section 3 discusses the empirical framework. Section 4 introduces the data. Section 5 presents estimation results and finally, Section 6 concludes.

2. Literature review

There is a growing body of literature on broadband diffusion with its main focus on the effects of competition and regulation on broadband penetration. In general, competition is found to have a significant and positive impact on broadband diffusion but with some differences with respect to the importance of inter- and intra-platform competition. Apart from competition and regulatory variables, some socioeconomic variables matter for broadband diffusion, such as GDP per capita, computer penetration, population density and education.

In linear regression for a panel data of 14 European countries between 2000 and 2004, Distaso, Lupi, and Manenti (2006) find that inter-platform competition has a positive impact on broadband diffusion while competition in the market for DSL services does not play a significant role. In another paper, using the logistic diffusion model for a panel data for U.S. Federal States between 1999 and 2004, Denni and Gruber (2007) find that both intra- and inter-platform competition are significant and positive but intra-platform competition has a positive impact only at the early stage of broadband diffusion. Lee and Brown (2008) estimate a logistic diffusion function using data for 30 OECD countries in the years 1999–2006 and find that competition between platforms has a positive impact on broadband diffusion at the early stage but the impact decreases when broadband market becomes mature. They also find that LLU regulation is statistically significant and increases broadband penetration.

Bouckaert et al. (2010) distinguish three types of competition between broadband Internet access providers that result from regulatory policies: (i) inter-platform competition which is not dependent on access regulation; (ii) facilities-based intra-platform competition which refers to the process where access seekers lease bare unbundled local loops and have to invest in equipment at some point in time; (iii) service-based intra-platform competition which offers two ways of access for the new entrants: bitstream or resale services offered by the incumbent. Their main research question is whether mandatory access facilitates broadband diffusion. They use a linear random effects regression model based on a panel of 20 OECD countries in the years 2003–2008 and find that inter-platform competition has a positive impact on broadband diffusion whereas the intra-platform competition variables have small positive impact or are insignificant.

Lee, Lee, and Marcu (2011) estimate separate logistic diffusion models for fixed and mobile broadband. Among other results, they find that local loop unbundling has a significant and positive impact on fixed broadband diffusion, while multiple standard policy has a significant and positive influence on mobile broadband diffusion. However, intra-platform competition is not significant in the fixed broadband regression. Finally, Gruber and Koutroumpis (2011) study the adoption of broadband services using quarterly panel data for 167 countries between 2000 and 2010. Similar to Lee et al. (2011), they do not find that inter-platform competition accelerates broadband adoption. Their results suggest that competition on the incumbent's DSL platform has a positive impact on broadband diffusion. They also find that unbundling of selected networks elements has a positive impact on the speed of diffusion. Finding no effect of inter-platform competition on broadband penetration the studies by Gruber and Koutroumpis (2011) and Lee et al. (2011) differ from the papers mentioned above, which reported positive effect of competition between platforms.

3. The data

Table 1 presents the list of variables used in the estimation and data sources. This paper differs from the previous studies because we use panel data for NUTS 1 regions in the EU in the years 2006–2010. In the regressions we use 385 observations for 96 out of 97 NUTS 1 regions. As reported in Table 2 there are many observations missing for broadband penetration. The data for other variables is relatively complete.

Broadband penetration: The dependant variable is broadband penetration measured as the number of broadband subscriptions per 100 households (*bb-pen*), which is published at NUTS 1 level by Eurostat.[5] We define broadband penetration at the household level rather than population level because it is still rather rare that a household has more than one broadband connection.

The following set of explanatory variables is used in the regression analysis:

Inter-platform Herfindahl index: To calculate the Herfindahl index for the broadband market (*hhi-inter*) we use the data from the "Eurobarometer: E-Communications Household Surveys" carried out by TNS Opinion & Social Network on behalf of the European Commission. The purpose of these surveys is to follow trends in electronic communications markets and to assess how EU households and citizens derive benefit from an increasingly competitive and innovative digital environment.[6] In this paper we use aggregate response data which is publicly available in the reports published on the website of the European Commission. The question asked to households which was the basis for the calculation of HHI index was "How does your household access the Internet from home?". Multiple answers were allowed with the following options: (i) ADSL; (ii) dial-up standard line; (iii) cable TV network; (iv) dial-up ISDN; (v) mobile phone network; and (vi) satellite network.[7] In the calculation of the Herfindahl index we considered only four options: DSL, cable modem, mobile broadband and other. Inter-platform competition is expected to have a positive impact on broadband adoption with the effect coming through lower prices and greater variety, quality and availability of broadband services. This variable is constructed on country-level due to lack of data on NUTS 1 level.

Prices for LLU: The prices for fully unbundled local loop connection (*llu-full*) and shared access connection (*llu-shared*) represent the cost of entry and provision of broadband services on the incumbent's network and therefore approximate intra-platform competition. The LLU prices are extracted from the "Progress Reports on the Single European Electronic Communications Market" which are published on the website of the European Commission on annual basis. These prices are reported on country-level.

Incumbent share in DSL connections: Another variable which approximates intra-platform competition is the incumbent's share in DSL connections (*dsl-share*) which is also extracted from the EC reports. A greater incumbent share in DSL connection can be expected to negatively influence broadband penetration because it may have a negative effect on broadband prices, product variety, quality and availability. This variable is reported on country-level.

In this analysis we cannot take into account the effect of any other regulatory variables such as the implementation of LLU because our data starts in 2006 when LLU was already mandated in almost all the EU countries. There is no sufficient variation to measure the effect of LLU implementation on broadband penetration other than via the effect of LLU prices.

Socioeconomic variables: The following socioeconomic variables are used in the regression analysis.

- Income (*gdp*) was found to be significant and positive in the previous studies on broadband penetration. It is approximated by GDP per capita (in 1000 Euros). This variable is used on NUTS 1 level and is available for the years 2006–2010 from Eurostat.
- Number of households (*num-house*) is available on NUTS 1 level and is taken from Eurostat regional market statistics database. The population of households consists of all households having at least one member of the age between 16 and 74 years.
- Computer penetration (*pc-pen*) measures the share of households having access to a computer. It is extracted from the Eurobarometer reports on country-level due to lack of information on NUTS 1 level.
- Population density (*density*) is available on NUTS 1 level from Eurostat.
- The education variable (*students*) measures the number of students and pupils who are in the ordinary school and university system relative to the total population size in the region. This data is available on NUTS 1 level from Eurostat.

[5] The data on broadband penetration is not available for the region of Bremen in Germany (DE5) at all. It is also not available for eight regions in the UK in 2010 for which we used an average for years 2009 and 2011. The other data is missing either at the beginning or at the end of the period and could not be interpolated.

[6] So far there were five Special Eurobarometer surveys focusing on telecommunications conducted in December 2005–January 2006; November–December 2006; November 2007–January 2008; November–December 2009 and February–March 2011. In the most recent survey, the interviews were conducted among 26,836 EU citizens in 27 Member States of the European Union. A technical note on the way in which the interviews were conducted by the institutes within the TNS Opinion & Social Network can be found in the annexes of the reports.

[7] The range of possible answers depends on the year of the survey. In the first survey conducted for the year 2006, only two answers were possible, either DSL or cable modem which were the main technologies at this time.

Table 1
Variables and data sources.

Description	Variable	Data source	Data level
Broadband penetration (% of households)	bb-pen	Eurostat (2006–2010), OECD	NUTS 1
HHI for broadband platforms	hhi-inter	EC Eurobarometer	Country
Prices per fully unbundled loop—connection (Euros)	llu-full	EC reports (2006–2010)	Country
Incumbent share in DSL connections	dsl-share	EC reports (2006–2010)	Country
GDP per capita (current prices)	gdp	Eurostat (2006–2010)	NUTS 1
Population density (population per km^2)	density	Eurostat (2006–2010)	NUTS 1
Number of households (in million)	num-house	Eurostat (2006–2010)	NUTS 1
Students and pupils as % of population	students	Eurostat (2006–2010)	NUTS 1
Computer penetration as (% of households)	pc-pen	EC Eurobarometer	Country
Years passed since introduction of broadband	years-intro	Czernich et al. (2011) & NRAs	Country

Table 2
Summary statistics.

Variable	*N*	Mean	Std. Dev.	Min	Max
bb	394	0.51	0.18	0.08	0.85
hhi	485	0.52	0.17	0.27	1.00
cable	485	0.18	0.15	0.00	0.62
dsl	485	0.58	0.23	0.05	1.00
dsl-share	485	0.64	0.19	0.31	1.00
llu-full	485	55.24	33.24	12.60	186.20
gdp	485	24.12	12.33	2.70	78.60
density	485	0.40	0.92	0.01	6.90
num-house	475	2.14	1.55	0.01	8.54
students	470	0.22	0.03	0.15	0.36
computer	485	0.58	0.14	0.13	0.95
years-intro	485	7.78	1.97	2.00	12.00

4. The empirical model

We analyze the determinants of broadband adoption using two different model specifications. First, we estimate a simple linear demand equation in which broadband penetration is regressed on prices for LLU connection, inter-platform HHI index, incumbent share in DSL connections and socioeconomic variables. Alternatively, we also estimate the same model in log–log specification. Next, we estimate the logistic diffusion function using the same set of explanatory variables.

4.1. Linear demand function

First, we estimate a simple linear reduced form regression in which broadband penetration, y_{it}, is regressed on demand factors denoted by D_{it} and supply factors which can be divided into cost factors denoted by C_{it} and markup determinants denoted by M_{it}, where i is country and t is time period subscript

$$y_{it} = D_{it}\alpha + C_{it}\beta + M_{it}\gamma + \epsilon_{it} \tag{1}$$

Demand factors include socioeconomic variables such as GDP per capita, share of students and pupils in the population, computer penetration. Cost factors include LLU prices and region-specific characteristics: population density and number of households which may represent economies of scale. Markup determinants include inter-platform HHI and incumbent share in DSL connections. The stochastic error term, ϵ_{it}, is assumed to be iid normally distributed. We also estimate log–log specification in which the penetration log is regressed on the same set of variables in logarithms.

4.2. Logistic diffusion

It is commonly observed that technologies follow an S-shaped diffusion pattern. After a slow take-up in the early diffusion stage, at some point of time technology becomes mainstream with exponential growth rates. Once it becomes widespread and approaches saturation level, the growth rate slows down again. Many empirical studies analyze determinants of diffusion of telecommunications services using the logistic diffusion model, see for instance Gruber and Verboven (2001), Denni and Gruber (2007), Lee and Brown (2008). The broadband diffusion pattern may be also approximated by an S-shaped function.

The logistic diffusion model can be specified as follows. Let y_{it} denote the number of consumers who have adopted broadband technology in country i at time t and let y_i^* denote the number of potential adopters in country i, i.e., the

maximum broadband penetration level. The diffusion curve is given by

$$y_{it} = \frac{y_i^*}{1 + \exp(-a_{it} - b_{it}t)} \tag{2}$$

where a_{it} is called the location or timing parameter which shifts the diffusion curve backward or forward depending on its sign but without changing the S-shape. The speed parameter b_{it} measures the speed of diffusion and represents the growth rate in the number of adopters relative to the number of agents who have not adopted the technology yet. After transformation we get the following linear equation to be estimated

$$\log\left(\frac{y_i^* - y_{it}}{y_{it}}\right) = -a_{it} - b_{it}t + \epsilon_{it} \tag{3}$$

Again the error term ϵ_{it} is assumed to be iid normally distributed.

We need to specify our three parameters: y_i^*, a_{it}, and b_{it}. The maximum number of adopters y_i^* can be defined in different ways. For instance, Czernich, Falck, Kretschmer, and Woessmann (2011) assume that the maximum reach of broadband is determined by the extension of the voice-telephony and cable-TV networks that existed before broadband. On the other hand, Gruber and Verboven (2001) assume that the number of potential adopters of mobile telephony varies proportionally to total population. We assume that the number of potential adopters is equivalent to the number of households since there will be in general at most one broadband connection per household.

As commonly specified in the previous studies, the speed of diffusion parameter b_{it} can be influenced by a greater degree of competition approximated in our model by inter-platform HHI index and incumbent share in DSL connections

$$b_{it} = \beta_0 + M_{it}\beta_1 \tag{4}$$

where M_{it} denotes markup determinants. The location parameter a_{it} can be specified as

$$a_{it} = \alpha_0 + D_{it}\alpha_1 + C_{it}\alpha_2 \tag{5}$$

where D_{it} denotes demand factors and C_{it} denotes cost determinants. For instance, demand for broadband is expected to increase with a higher level of income, a higher share of pupils and students, and a higher penetration rate of computers. On the supply side, a higher population density may decrease the costs of deployment and increase broadband penetration via lower prices. There may be also economies of scale in broadband supply since a greater number of households in a local market can lower costs and prices leading to higher penetration. LLU prices also influence the cost of entrants and may influence broadband penetration.

As suggested by some of the previous studies, there is a potential endogeneity issue of the HHI inter-platform index, see Distaso et al. (2006) and Bouckaert et al. (2010). A higher concentration of broadband market suggests that there is little competition which may result in lower broadband penetration. However, an unobserved positive shock to demand for broadband may have a positive impact on entry of different broadband platforms and the HHI index will decrease. Hence, an OLS regression may overestimate the effect of the HHI index on market penetration. The argumentation as to why incumbent share in DSL connections may be endogenous is similar. As mentioned by Bouckaert et al. (2010), the endogeneity issue may be mitigated to some extent by inclusion of fixed or random effects because the unobserved region-specific factors are then filtered out.

5. Estimation results

Tables 3 and 4 present the estimation results for Eqs. (1) and (3), respectively. We estimate these equations using both random and fixed effects regressions. These models control for all time-invariant differences between regions absorbing the impact of variables which do not vary significantly over time. The fixed effects regression allows for the fixed effects to be correlated with the independent variables. The random effects regression, on the other hand, takes into account the presence of region-specific effects which are assumed to be uncorrelated with the independent variables. If the assumption on the random effects holds, the random effects model is more efficient than the fixed effects model. But if this assumption does not hold, the random effects model is not consistent. To decide between the fixed and random effects models we use the Hausman test, in which the null hypothesis is that the unique errors are not correlated with the regressors, i.e., the preferred model is the random effects model.

5.1. Linear demand function

First, we estimate linear demand specification, the results of which are shown in Table 3. Broadband penetration is regressed on prices for a fully unbundled loop connection, inter-platform HHI and incumbent share in DSL connections.[8] The following socioeconomic variables are included to control for observed heterogeneity across regions due to differences in demand and cost factors: GDP per capita (in 1000 Euros on NUTS 1 level), population density (in 1000 per km^2 on NUTS 1

[8] As an alternative specification we used prices for shared access connection which are also significant with a negative coefficient in all the specifications.

level), number of households (in 1000 on NUTS 1 level), pupils and students (as % of population on NUTS 1 level), and share of households with computers (as % on country level). In addition, we include a time trend variable represented by the number of years passed since the launch of broadband in the country. We control for unobserved heterogeneity by estimating both random and fixed effects models. In the case of linear specification, the Hausman test with statistics $\chi^2 = 52.9$ allows us to reject the null hypothesis that the random effects are not correlated with the regressors with a significance level of 0.001. Also, in the case of log–log specification, the Hausman test with statistics $\chi^2 = 36.9$ allows to reject the null hypothesis with a significance level of 0.001. Hence, for both specifications the fixed effects model is preferred to the random effects model.

The estimation results are as follows. In all the regressions, HHI is found to be significant and negative which suggests that a higher level of inter-platform concentration, i.e., less inter-platform competition, has a negative effect on broadband penetration. This result is consistent with findings in most of the previous studies. In general, HHI takes higher values in countries in which DSL broadband has a higher share due to country-specific policies towards broadband and because of historical reasons. In Western Europe for instance, DSL has been the most common technology for Internet access based on the telephone network developed before liberalization of telecommunications markets in 1998. On the other hand, in the post-socialist countries in Central and Eastern Europe, where telecommunications infrastructure inherited from the socialism era was in a poor state before liberalization in the early 2000s, alternative technologies were rolled out and taken up more swiftly than in the old Member States. The estimation results suggest that after controlling for differences in GDP per capita and other socioeconomic variables, broadband penetration tends to be higher in countries in which alternative technologies to DSL have significant market shares. As an additional test, we estimate two alternative regressions in which HHI is replaced by market share of DSL and by market share of cable modem. The impact of DSL market share is significant and negative and the impact of cable modem market share is significant and positive, which confirms that inter-platform competition stimulates broadband penetration.

Apart from inter-platform competition, also more competition on DSL platform has a significant and positive impact on broadband penetration, except fixed effects linear specification. The competition on DSL platform is approximated by two variables. First, LLU prices represent the cost of entry and provision of broadband services on the incumbent's DSL network. We find that lower prices for a fully unbundled local loop connection have a significant and positive impact on broadband penetration in all the regressions. In general, lower LLU prices may encourage entry and imply lower costs for the entrants enabling them to charge lower prices. Second, we find that incumbent share in DSL connections has a negative impact on broadband penetration. As discussed above, the lower level of intra-platform competition in the new EU countries may be offset by a higher level of inter-platform competition.

Some socioeconomic variables are significant in explaining heterogeneity across regions depending on the specification. GDP per capita which approximates the income effect commonly found in the previous studies is significant and positive only in the random effects models. Density of population is not significant except the random effects estimation of the log–log specification, in which it is significant and positive. This is a proxy for the cost of deployment of broadband because it is less expensive to provide broadband services in small geographic areas which are densely populated. The number of households is not significant in both linear and log–log specifications. This suggests lack of economies of scale and local market size effect. The share of pupils and students is statistically significant with a negative impact on broadband adoption in the fixed effects regressions and insignificant in the random effects regressions. This result is rather counterintuitive because we would expect a greater demand for broadband in areas with a higher share of pupils and students. Broadband penetration is on the other hand higher in countries with a higher share of households having computers at home as expected. The computer penetration is significant and positive in all the regressions except the fixed effects linear demand specification. Time trend variable is significant and positive in all the specifications due to the fact that broadband penetration has been continuously growing in the last years.

We explain broadband penetration on NUTS 1 level using a mix of socioeconomic variables on NUTS 1 level and country-level competition and regulation variables. In general, we may expect more inter-platform competition in local markets with greater population size, higher density of population and higher GDP per capita, which are also markets with higher broadband penetration. The impact of inter-platform HHI on broadband penetration should be higher for NUTS 1 level data than for country-level data. The same argumentation holds for the incumbent share in DSL connections. To verify this we estimate additional regression using country-level broadband data in years 2006–2010 with the same set of explanatory variables. The number of observations is reduced to 133 for 27 EU countries. In this regression, the coefficient on HHI index of −0.139 in fixed effects regression (−0.154 in random effects regression) is smaller than in NUTS 1 level fixed effects regressions of −0.148 (−0.184 in random effects regression) from Table 3. The incumbent's share in DSL connections is insignificant. The insignificance may be due to the fact that this regression uses a smaller number of observations and there is less variation in data on broadband penetration.

5.2. Logistic diffusion

The estimation of the logistic diffusion function yields results which are comparable to linear demand estimation, as shown in Table 4. The dependant variable is the logarithm of the number of potential adopters who do not have broadband yet relative to the maximum number of adopters given by Eq. (3). We estimate both fixed and random effects models. The Hausman test with statistics $\chi^2 = 66.1$ allows us to reject the null hypothesis that the random effects are not correlated with

Table 3
Estimation results for penetration regression.

Variables	Linear specification		Variables	Log–log specification	
	FE	RE		FE	RE
gdp	−0.0001	0.0019***	log(gdp)	0.159	0.121***
	(*−0.49*)	(*2.68*)		(*1.47*)	(*3.02*)
density	−0.040	0.006	log(density)	−0.015	0.029
	(*−1.11*)	(0.67)		(*−0.39*)	(*1.69*)
num-house	0.044	−0.002	log(num-house)	0.104	−0.027
	(*1.55*)	(*−0.48*)		(*1.34*)	(*−1.31*)
students	−1.915***	0.289	log(students)	−1.636***	0.126
	(*−2.66*)	(*1.08*)		(*−3.77*)	(*0.86*)
pc-pen	0.001	−0.395***	log(pc-pen)	0.409***	0.612***
	(*0.99*)	(*6.31*)		(*4.83*)	(*8.57*)
years-intro	0.061***	0.046***	years-intro	0.105***	0.094***
	(*14.58*)	(*13.67*)		(*11.69*)	(*12.14*)
hhi-inter	−0.148***	−0.184***	log(hhi-inter)	−0.251***	−0.317***
	(*−6.42*)	(*−8.60*)		(*−6.07*)	(*−8.95*)
llu-full	−0.0004***	−0.0005***	log(llu-full)	−0.051***	−0.069***
	(*−3.24*)	(*−4.44*)		(*−2.38*)	(*−3.77*)
dsl-share	−0.114	−0.155***	log(dsl-share)	−0.494***	−0.281***
	(*−1.54*)	(*−3.94*)		(*−3.37*)	(*−4.21*)
Intercept	0.481***	0.006	Intercept	−4.732***	−1.417***
	(*2.85*)	(*0.93*)		(*−6.28*)	(*−4.78*)
Observations	385	385		385	385
Groups	96	96		96	96

t statistics are in parentheses. Variables in the log–log demand specification are all expressed in logarithm except the variable "years-intro" which represents time trend.
* Significance at 10% level.
** Significance at 5% level.
*** Significance at 1% level.

the regressors with a significance level of 0.001. Hence, the fixed effects model is preferred to the random effects model. Apart from the dependent variable, the difference between the linear demand model and the logistic diffusion function is that socioeconomic factors and LLU prices are assumed to influence the location parameter, i.e., broadband penetration at a given point of time, while inter-platform Herfindahl index, market shares of different broadband technologies and incumbent share in DSL connections are assumed to influence the diffusion speed parameter. This is the common way in which competition variables were introduced in diffusion function estimation in the previous studies.

In both fixed and random effects regressions, HHI is found to be negative and significant which suggests that a higher level of platform concentration has a negative effect on the speed of broadband diffusion. Again, we estimate two alternative regressions in which HHI is replaced by market share of DSL and cable modem. Broadband diffusion is slower in countries with a higher DSL market share which is found to be significant and negative, while the impact of the share of cable modem is significant and positive. This confirms that inter-platform competition stimulates the speed of broadband diffusion. Also, incumbent share in DSL connections has a significant and negative impact on broadband penetration in both fixed and random effects regressions. Thus, both inter- and intra-platform competition accelerates broadband diffusion. The LLU prices are however found to be insignificant.

Some socioeconomic variables are significant in explaining the heterogeneity across regions with respect to broadband penetration depending on the specification. GDP per capita is significant in both fixed and random effects regressions. Density of population is not significant. The number of households has a significant and positive impact on broadband diffusion in the fixed effects regression but it is not significant in the random effects regression. The share of pupils and students is statistically significant with a negative effect on broadband diffusion in the fixed effects regression but it is insignificant in the random effects regression. Broadband penetration is higher in countries with a higher share of households having computers at home.

6. Conclusions

Universal access to broadband is critical for the creation of the information society and economic progress. It is therefore important to analyze the determinants of broadband penetration with a particular focus on inter- and intra-platform competition and regulation in order to provide guidance on government policies towards deployment of broadband.

In general, the previous studies analyzed determinants of broadband diffusion using country-level panel data and found that inter-platform competition stimulates broadband diffusion, while intra-platform competition has a positive impact on

Table 4
Estimation results for logistic regression.

Variables	FE	RE
Location variables		
gdp	0.027***	0.022***
	(*3.66*)	(*−6.08*)
density	−0.146	−0.026
	(*0.91*)	(*0.55*)
num-house	0.337***	0.005
	(*−3.02*)	(*0.21*)
students	−11.16***	1.601
	(*3.66*)	(*−1.24*)
pc-pen	0.634*	2.221***
	(*−1.64*)	(*−8.15*)
llu-full	−0.0001	−0.001
	(*1.29*)	(*1.18*)
Intercept	−0.261	−2.891***
	(*0.33*)	(*9.99*)
Diffusion variables		
Time*hhi-inter	−0.133***	−0.183***
	(*3.35*)	(*4.86*)
Time*dsl-share	−0.241***	−0.270***
	(*5.48*)	(*7.74*)
Time	0.485***	0.481***
	(*13.39*)	(*−13.99*)
Observations	385	385
Groups	96	96

t statistics are in parentheses.
* Significance at 10% level.
** Significance at 5% level.
*** Significance at 1% level.

broadband penetration in the short run, if at all. But there are also studies which found that competition on the incumbent's DSL platform has a positive impact on broadband diffusion, while inter-platform competition does not matter at all.

In this paper, we use data for 96 NUTS 1 regions in the EU to analyze the impact of competition and regulation on broadband penetration. The advantage of our paper is that we use more detailed data than country-level which allows us to control for regional heterogeneity more accurately. We include in the regression analysis socioeconomic variables to control for observable heterogeneity between regions and account for unobserved heterogeneity by estimating fixed and random effects models. Our results suggest that policies promoting both inter- and intra-platform competition matter for broadband diffusion. We find that a higher value of inter-platform HHI results in lower broadband penetration. Broadband deployment is lower in countries in which DSL has a greater share in Internet access and higher in countries in which cable modem has a greater share in Internet access. We also find that a higher incumbent share in DSL connections decreases broadband penetration. Furthermore, prices for a fully unbundled local loop connection, which represent the cost of providing copper-based Internet services, have a significant and negative impact on broadband penetration.

Acknowledgment

We gratefully acknowledge the financial support from Telecom Paris Tech and the Economic Research Southern Africa.

References

Bouckaert, J., van Dijk, T., & Verboven, F. (2010). Access regulation, competition, and broadband penetration: An international study. *Telecommunications Policy*, *34*, 661–671.
Czernich, N., Falck, O., Kretschmer, T., & Woessmann, L. (2011). Broadband infrastructure and economic growth. *Economic Journal*, *121*, 505–532.
Denni, M., & Gruber, H. (2007). The diffusion of broadband telecommunications in the U.S.—the role of different forms of competition. *Communications and Strategies*, *68*, 139.
Distaso, W., Lupi, P., & Manenti, F. M. (2006). Platform competition and broadband uptake: Theory and empirical evidence from the European Union. *Information Economics and Policy*, *18*, 87–106.
Gruber, H., & Koutroumpis, P. (2011). *Competition enhancing regulation and diffusion of innovation: The case of broadband networks*. Working Paper.
Gruber, H., & Verboven, F. (2001). The diffusion of mobile telecommunications services in the European Union. *European Economic Review*, *45*, 577–588.
Lee, S., & Brown, J. S. (2008). *The diffusion of fixed broadband: An empirical analysis* (pp. 08–19). NET Institute Working Paper.
Lee, S., Lee, S., & Marcu, M. (2011). An empirical analysis of fixed and mobile broadband diffusion. *Information Economics and Policy*, *23*, 227–233.